Evaluate Yourself

1. What are modifiers? What is the use of modifiers?
2. What is wrong with the following C++ statement:
   long float x;
3. What is a variable ? Why is a variable called symblolic variable?
4. What do you mean by dynamic initialization of a variable? Give an exmple.
5. What is wrong with the following statement?
   const int x;

9.14 Formatting Output

Formatting output is very important in the development of output screens for easy reading and understanding. Manipulators are used to format the output of any C++ program. Manipulators are functions specifically designed to use with the insertion (<<) and extraction(>>) operators.

C++ offers several input and output manipulators for formatting. Commonly used manipulators are: **endl, setw, setfill, setprecision and setf**. In order to use these manipulators, you should include the appropriate header file. **endl** manipulator is a member of iostream header file. **setw, setfill, setprecision and setf** manipulators are members of iomanip header file.

endl (End the Line)

endl is used as a line feeder in C++. It can be used as an alternate to '\n'. In other words, endl inserts a new line and then makes the cursor to point to the beginning of the next line. There is a difference between endl and '\n', even though they are performing similar tasks.

- endl – Inserts a new line and flushes the buffer (Flush means – clean)
- '\n' - Inserts only a new line.

Example:

```
cout << "\n The value of num = " << num;
cout << "The value of num = " << num <<endl;
```

Both these statements display the same output.

setw ( )

setw manipulator sets the **width of the field** assigned for the output. The field width determines the minimum number of characters to be written in output.

Syntax:

setw(number of characters)

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